

How to Choose the Right Open Source Database

Databases store data in an organised manner so that its retrieval becomes easy. Also, the management of data is easier when it is stored in a database. There are many factors to be considered, however, before choosing a database for a particular software application. Let's take a look at how we can make the right choice.



Considering the number of users working with different online applications nowadays, databases are about the most important part of any software application, and can make a huge impact on its performance. It is really difficult and time consuming for a database to operate if it has a huge load of data or if it has a variety of data sets to operate upon. There are also several other factors like the security of data, as well as the cost (if there is one) associated with the database, impacting the choice of the database. A free database helps users to avoid huge costs. Hence, open source databases have been playing quite an important role in many of the widely used software applications across the globe.

Different kinds of databases available in the market

There are various types of databases being used by different users across the globe, based on their requirement. The following is a broad classification of databases.

1. **Relational databases:** Relational databases are the most common among all the types. In such databases, the data is actually stored in the form of different data tables. Each of the tables has a unique key field and that key is used to connect one table to the other tables. Hence, different tables are related to each other with the help of various key fields. Such databases are widely used in industries
2. **Operational databases:** An operational database is very important for organisations, as it supports the customer database and the inventory database. It helps companies to keep a track of inventories as well as store details of the customers who buy its products. The data stored in various operational databases can be analysed and used based on the requirements of the company.
3. **Database warehouses:** There might be a requirement for different organisations to keep some of the relevant data for several years. Such data sets act as significant sources of information to compare and analyse the present year's data with that of previous years, which makes it quite easy to determine key trends. All such data from previous years is maintained and stored in a large database warehouse. Since the stored data set has already gone through various kinds of editing, screening and integration, it no longer requires any editing or alteration.
4. **Distributed databases:** Many organisations have several regional offices, manufacturing plants, branch offices and a head office. Each such work group may have its own set of databases, which collectively form the main database of the company. Such a system of databases is known as a

like media, telecom, etc, and this is probably the type of database one is most likely to come across.